

(4.1)

Hledáme koeficienty $\hat{a}, \hat{b}, \hat{c}$, které nejlepším způsobem vyhovují rovnici $A\hat{x} = r$.

$$r = ax + by + c$$

Body:

$$\bullet 0 \cdot a + 0 \cdot b + c = 1$$

$$\bullet 1 \cdot a + 0 \cdot b + c = 1$$

$$\bullet 0 \cdot a + 1 \cdot b + c = 1$$

$$\bullet 1 \cdot a + 1 \cdot b + c = 2$$

$$\bullet 1 \cdot a + (-1) \cdot b + c = 0$$

$$\bullet -1 \cdot a + 1 \cdot b + c = 1$$

$$A = \begin{matrix} & \hat{a} & \hat{b} & \hat{c} \\ \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} & r = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} \end{matrix}$$

$$A^T A \hat{x} = A^T r$$

$$A^T A = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} = G$$

$$A^T r = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = u$$

Řešíme rovnici $G\hat{x} = u$:

$$\left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 4 & -1 & 2 & 2 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 15 & 10 & 18 \\ 0 & 10 & 10 & 14 \end{array} \right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 5 & 5 & 7 \\ 0 & 15 & 10 & 18 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & -5 & -3 \end{array} \right) \Rightarrow -5\hat{c} = -3 \Rightarrow \underline{\hat{c} = \frac{3}{5}}$$

$$5\hat{b} + 5\hat{c} = 7$$

$$5\hat{b} + 3 = 7$$

$$5\hat{b} = 4 \Rightarrow \underline{\hat{b} = \frac{4}{5}}$$

$$-\hat{a} + 4\hat{b} + 2\hat{c} = 4$$

$$-\hat{a} + \frac{16}{5} + \frac{6}{5} = 4$$

$$-\hat{a} = 4 - \frac{22}{5}$$

$$\underline{\hat{a} = \frac{22}{5} - 4 = \frac{22 - 20}{5} = \frac{2}{5}}$$

$$\boxed{r = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}}$$